

The possible total points are 66:

P1(20): (a) the scheme worth 3 points and DOF worth 3 points; (b) 4 points; (c) 4 points; (d) 6 points; you can use a solver embedded in any computational tools to solve; be reminded for non-linear systems, Newton iteration heavily relies on the initial guess and there exist multiple local minimums, therefore you should try many times and make sure the solution is in the bound. Presenting a plot of T vs yield or getting the optimum worth 3 points.

P2(15): There are two different ways to insure full marks: 1. solve the problem exactly the same as 2.69 but this time a solution needs be presented which worth 5 points; 2. introduce a factor α representing the ratio of the flow back to the mixer from the splitter; you don't need solve the equation system for lack of one constraint, but need explain how the factor affects the system.

P3(15): As long as a discrepancy between the inputs and the measured values from the chemist is located, you can state the chemist is incorrect. The statement is essential and values 3 points

P4(16): Mole fractions for CO₂ and CO each worth 4 points; the rest of the compositions worth 2 points each.